PROBLEM 16-11 Toluene is more reactive; the trifluoromethyl group is electron-withdrawing.

PROBLEM 16-12 The nitrogen electrons are donated to the nearby carbonyl group by resonance and are less available to the ring.

PROBLEM 16-13 The meta intermediate is most favored.

PROBLEM 16-14 **(a)** Ortho and para to $-OCH_3$ **(b)** Ortho and para to $-NH_2$ **(c)** Ortho and para to $-Cl$

PROBLEM 16-15 **(a)** Reaction occurs ortho and para to the $-CH_3$ group.
(b) Reaction occurs ortho and para to the $-OCH_3$ group.

PROBLEM 16-16 The phenol is deprotonated by KOH to give an anion that carries out a nucleophilic acyl substitution reaction on the fluoronitrobenzene.

PROBLEM 16-17 Only one benzyne intermediate can form from *p*-bromotoluene; two different benzyne intermediates can form from *m*-bromotoluene.

PROBLEM 16-18 **(a)** *m*-Nitrobenzoic acid **(b)** *p*-*tert*-Butylbenzoic acid

PROBLEM 16-19 A benzyl radical is more stable than a primary alkyl radical by 52 kJ/mol and is similar in stability to an allyl radical.

PROBLEM 16-20 1. CH_3CH_2Cl, $AlCl_3$; 2. NBS; 3. KOH, ethanol

PROBLEM 16-21 1. PhCOCl, $AlCl_3$; 2. H_2/Pd

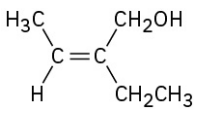
PROBLEM 16-22 **(a)** 1. HNO_3, H_2SO_4; 2. Cl_2, $FeCl_3$ **(b)** 1. CH_3COCl, $AlCl_3$; 2. Cl_2, $FeCl_3$; 3. H_2/Pd
(c) 1. CH_3CH_2COCl, $AlCl_3$; 2. Cl_2, $FeCl_3$; 3. H_2/Pd; 4. HNO_3, H_2SO_4
(d) 1. CH_3Cl, $AlCl_3$; 2. Br_2, $FeBr_3$; 3. SO_3, H_2SO_4

PROBLEM 16-23 **(a)** Friedel–Crafts acylation does not occur on a deactivated ring.
(b) Rearrangement occurs during Friedel–Crafts alkylation with primary halides; chlorination occurs ortho to the alkyl group.

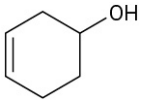
Chapter 17

PROBLEM 17-1 **(a)** 5-Methyl-2,4-hexanediol **(b)** 2-Methyl-4-phenyl-2-butanol **(c)** 4,4-Dimethylcyclohexanol **(d)** *trans*-2-Bromocyclopentanol **(e)** 4-Bromo-3-methylphenol **(f)** 2-Cyclopenten-1-ol

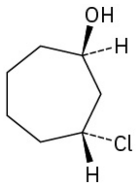
PROBLEM 17-2 **(a)**



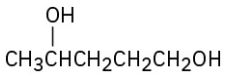
(b)



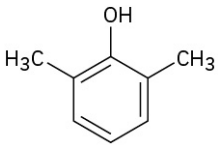
(c)



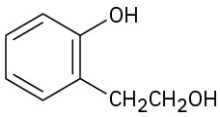
(d)



(e)



(f)



PROBLEM 17-3 Hydrogen-bonding is more difficult in hindered alcohols.

PROBLEM 17-4 **(a)** $HC{\equiv}CH < (CH_3)_2CHOH < CH_3OH < (CF_3)_2CHOH$
(b) *p*-Methylphenol < Phenol < *p*-(Trifluoromethyl)phenol
(c) Benzyl alcohol < Phenol < *p*-Hydroxybenzoic acid

PROBLEM 17-5 The electron-withdrawing nitro group stabilizes an alkoxide ion, but the electron-donating methoxyl group destabilizes the anion.

PROBLEM 17-6 **(a)** 2-Methyl-3-pentanol **(b)** 2-Methyl-4-phenyl-2-butanol **(c)** *meso*-5,6-Decanediol

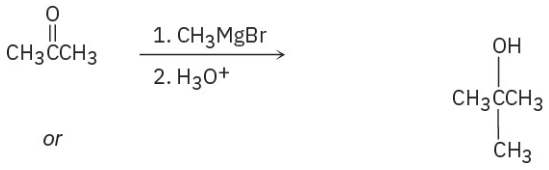
PROBLEM **(a)** $NaBH_4$ **(b)** $LiAlH_4$ **(c)** $LiAlH_4$

17-7

PROBLEM 17-8 **(a)** Benzaldehyde or benzoic acid (or ester) **(b)** Acetophenone **(c)** Cyclohexanone **(d)** 2-Methylpropanal or 2-methylpropanoic acid (or ester)

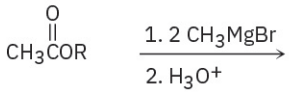
PROBLEM 17-9 **(a)** 1-Methylcyclopentanol **(b)** 1,1-Diphenylethanol **(c)** 3-Methyl-3-hexanol

PROBLEM 17-10 **(a)**

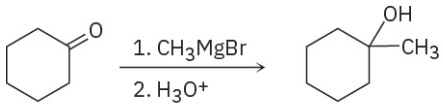


or

2-Methyl-2-propanol

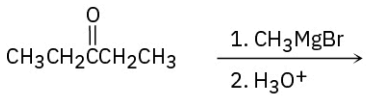


(b)

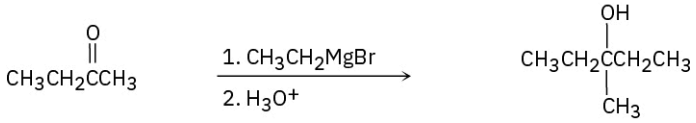


1-Methylcyclohexanol

(c)

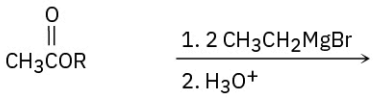


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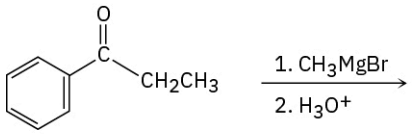


3-Methyl-3-pentanol

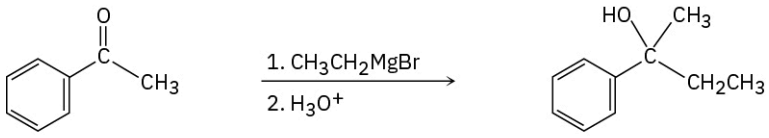
or



(d)

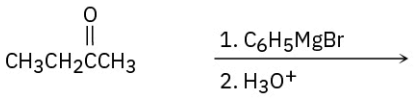


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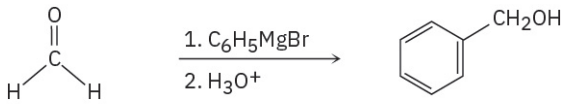


2-Phenyl-2-butanol

or

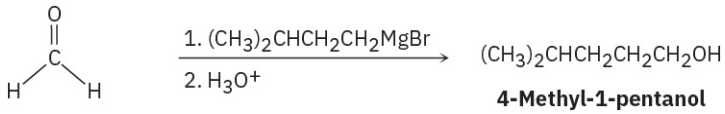


(e)



Benzyl alcohol

(f)



4-Methyl-1-pentanol

PROBLEM 17-11 Cyclohexanone + CH_3CH_2MgBr

PROBLEM 17-12 1. *p*-TosCl, pyridine; 2. NaCN

PROBLEM 17-13 **(a)** 2-Methyl-2-pentene **(b)** 3-Methylcyclohexene **(c)** 1-Methylcyclohexene **(d)** 2,3-Dimethyl-2-pentene **(e)** 2-Methyl-2-pentene

PROBLEM 17-14 **(a)** 1-Phenylethanol **(b)** 2-Methyl-1-propanol **(c)** Cyclopentanol

PROBLEM 17-15 **(a)** Hexanoic acid, hexanal **(b)** 2-Hexanone **(c)** Hexanoic acid, no reaction

PROBLEM 17-16 S_N2 reaction of F^- on silicon with displacement of alkoxide ion.

PROBLEM 17-17 Protonation of 2-methylpropene gives the *tert*-butyl cation, which carries out an electrophilic aromatic substitution reaction.

PROBLEM 17-18 Disappearance of –OH absorption; appearance of C=O

PROBLEM 17-19 **(a)** Singlet **(b)** Doublet **(c)** Triplet **(d)** Doublet **(e)** Doublet **(f)** Singlet

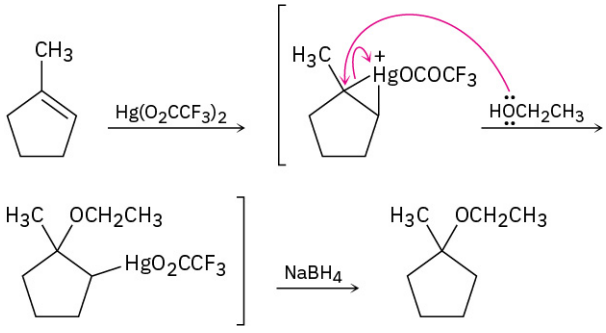
Chapter 18

PROBLEM 18-1 **(a)** Diisopropyl ether **(b)** Cyclopentyl propyl ether **(c)** *p*-Bromoanisole or 4-bromo-1-methoxybenzene **(d)** 1-Methoxycyclohexene **(e)** Ethyl isobutyl ether **(f)** Allyl vinyl ether

PROBLEM 18-2 A mixture of diethyl ether, dipropyl ether, and ethyl propyl ether is formed in a 1 : 1 : 2 ratio.

PROBLEM 18-3 **(a)** $CH_3CH_2CH_2O^-$ + CH_3Br **(b)** PhO^- + CH_3Br **(c)** $(CH_3)_2CHO^-$ + $PhCH_2Br$ **(d)** $(CH_3)_3CCH_2O^-$ + CH_3CH_2Br

PROBLEM 18-4



PROBLEM 18-5 **(a)** Either method **(b)** Williamson **(c)** Alkoxymercuration **(d)** Williamson

PROBLEM 18-6 **(a)** Bromoethane > 2-Bromopropane > Bromobenzene **(b)** Bromoethane > Chloroethane > 1-Iodopropene